1. MC.07.01.ALGO (Algorithmic)

If the variable cost per unit goes down,

a. contribution margin decreases and break-even point increases.

✓ b. contribution margin increases and break-even point decreases.

c. fixed cost increases and break-even point increases.

d. contribution margin decreases and break-even point remains unchanged.

e. total variable cost increases and break-even point decreases.

Feedback

Check My Work

Set up a simple CVP example and calculate the contribution margin and break-even units. Reduce the variable cost per unit and calculate the new contribution margin and break-even units.

Solution
b

2. MC.07.02.ALGO (Algorithmic)

The amount of revenue required to earn a targeted profit is equal to

a. targeted profit divided by sales price per unit.

b. total fixed cost divided by the contribution margin ratio.

✓ c. total fixed cost plus targeted profit divided by contribution margin ratio.

d. targeted profit divided by the variable cost ratio.

e. total fixed cost plus total variable cost divided by contribution margin.

Feedback

Check My Work

Break-even analysis can be adjusted to focus on targeted profit.

Solution
c

3. MC.07.03.ALGO (Algorithmic)

Break-even revenue for a multiple-product firm can

✓ a. be calculated by dividing total fixed cost by the overall contribution margin ratio.

b. be calculated by dividing total fixed cost by the overall variable cost ratio.

c. be calculated by dividing segment fixed cost by unit sales per segment.

d. be calculated by multiplying total fixed cost by the contribution margin ratio.

e. not be calculated, break-even revenue can only be computed for a single-product firm.

Feedback

Check My Work

Consider the relationship between revenue and contribution margin.

Solution

a

4. MC.07.04

In the cost-volume-profit graph,

a. the break-even point is found where the total revenue curve crosses the x-axis.

b. the area of profit is to the left of the break-even point.

c. the area of loss cannot be determined.

✓ d. both the total revenue curve and the total cost curve appear.

e. neither the total revenue curve nor the total cost curve appears.

Feedback

Check My Work

Consider the components of the CVP graph.

Solution

d

5. MC.07.05

An important assumption of cost-volume-profit analysis is that

a. both costs and revenues are linear functions.

b. all cost and revenue relationships are analyzed within the relevant range.

c. there is no change in inventories.

d. the sales mix remains constant.

✓ e. all of these are assumptions of cost-volume-profit analysis.

Feedback

Check My Work

Review assumptions of CVP analysis.

Solution

e

6. MC.07.06.ALGO (Algorithmic)

The use of fixed costs to extract higher percentage changes in profits as sales activity changes involves

✓ a. operating leverage.

b. financial leverage.

c. sensitivity analysis.

d. break-even analysis.

e. All of these.

Feedback

Check My Work

Review the chapter's key terms.

Solution

a

7. MC.07.07.ALGO (Algorithmic)

If the margin of safety is 0, then

a. the company is earning its targeted profit.

b. the company is not operating at break-even revenue.

c. the margin of safety cannot be less than or equal to 0; it must be positive.

✓ d. the company is precisely breaking even.

e. the company's total variable costs are equal to its total fixed costs.

Feedback

Check My Work

Review the definition of *margin of safety*.

Solution

d

8. MC.07.08

The contribution margin is the

a. amount by which sales exceed total fixed cost.

b. difference between sales and total cost.

c. difference between sales and operating income.

✓ d. difference between sales and total variable cost.

e. difference between variable cost and fixed cost.

Feedback

Check My Work

Review the chapter's key terms.

Solution

d

9. MC.07.09.ALGO (Algorithmic)

Dartmouth Company produces a single product with a price of $8, variable cost per unit of $4, and total fixed cost of $8,000. Dartmouth's break-even point in units **(round your answer to the nearest whole number)**

a. is 2,016.

b. is 2,008.

c. is 1,992.

✓ d. is 2,000.

e. cannot be determined from the information given.

Feedback

Check My Work

Use the operating income equation with the given numbers.

Post-Submission

Break-Even Units = $8,000/($8 − $4) = 2,000

Solution
d

10. MC.07.10.ALGO (Algorithmic)

Dartmouth Company produces a single product with a price of $12, variable cost per unit of $3, and total fixed cost of $8,900.

The variable cost ratio and the contribution margin ratio for Dartmouth, **rounded to the nearest whole number,** are

a. 25% and 25%, respectively.

b. 75% and 25%, respectively.

c. 75% and 75%, respectively.

✓ d. 25% and 75%, respectively.

e. The contribution margin ratio cannot be determined from the information given.

Feedback

Check My Work

Use the formulas for variable cost ratio and contribution margin ratio.

Post-Submission

Variable Cost Ratio = $3/$12 = 0.25, or 25%

Contribution Margin Ratio = ($12 − $3)/$12 = 0.75, or 75%

Solution
d

11. MC.07.11

If a company's total fixed cost decreases by $10,000, which of the following will be true?

a. The break-even point will increase.

b. The variable cost ratio will increase.

c. The break-even point will be unchanged.

✓ d. The variable cost ratio will be unchanged.

e. The contribution margin ratio will increase.

Feedback

Check My Work

Consider the relationship between fixed costs and the other elements of the contribution margin income statement.

Solution
d

12. MC.07.12

Solemon Company has total fixed cost of $15,000, variable cost per unit of $6, and a price of $8. If Solemon wants to earn a targeted profit of $3,600, how many units must be sold?

a. 2,500

b. 7,500

c. 9,300

d. 18,600

e. 18,750

Feedback

Check My Work

Calculate the contribution margin per unit. Add targeted income and fixed costs then divide by the contribution margin per unit.

Post-Submission

Units to Be Sold = ($15,000 + $3,600)/($8 − $6) = 9,300

Solution
c

13. MC.07.13

Which of the following is a use of data analytics in cost-volume-profit analysis?

a. Developing a cost-volume-profit graph.

b. Performing sensitivity analysis on variable costs.

c. Performing sensitivity analysis on fixed costs.

d. Calculating the break-even point using budgeted, not actual, data.

e. All of these are applications of data analytics.

Feedback

Check My Work
Correct

Solution
e

14. PR.07.161.ALGO (Algorithmic)

A dance academy plans to sell 11,300 ballet shoes at $65 each in the coming year. The per-unit variable cost is $30, and the total fixed cost equals $71,900.

Required:

A. Calculate the break-even number of ballet shoes. **Note**: **Round units Up to next whole unit. Use rounded amount in subsequent computations.**

2055

ballet shoes

B. Calculate the break-even sales dollars.

$

133575

Feedback

Check My Work

Correct

Post-Submission

A.

$71,900 ÷ ($65 - $30) = 2,055 ballet shoes

B.

2,055 ballet shoes × $65 = $133,575

OR

$71,900 ÷ [($65 - $30) ÷ $65] = $133,575

Solution

A dance academy plans to sell 11,300 ballet shoes at $65 each in the coming year. The per-unit variable cost is $30, and the total fixed cost equals $71,900.

Required:

A. Calculate the break-even number of ballet shoes. **Note**: **Round units Up to next whole unit. Use rounded amount in subsequent computations.**

2055 ballet shoes

B. Calculate the break-even sales dollars.

$ 133575

15. PR.07.163.ALGO (Algorithmic)

Nolan Company expects to produce and sell 2,000 units in the next month. The data on costs are as follows:

Per-unit costs:	
Selling price	$49
Variable manufacturing costs	10
Variable selling costs	5
Total costs:	
Fixed manufacturing costs	$19,200
Fixed selling costs	7,100

Required:

A. What is the variable cost per unit?

$ ✓

15

B. What is the contribution margin per unit?

$ ✓

34

C. What is the variable cost ratio?

✓

31

%

D. What is the contribution margin ratio?

✓

69

%

Feedback

Check My Work

Correct

Post-Submission

A.

Variable Cost per Unit = $10 + $5 = $15

B.

Contribution Margin per Unit = $49 - $15 = $34

C.

Variable Cost Ratio = $15 ÷ $49 = 0.31, or 31%

D.

Contribution Margin Ratio = $34 ÷ $49 = 0.69 or 69% OR 1 - 31% = 69%

Solution

Nolan Company expects to produce and sell 2,000 units in the next month. The data on costs are as follows:

Per-unit costs:	
Selling price	$49
Variable manufacturing costs	10
Variable selling costs	5
Total costs:	
Fixed manufacturing costs	$19,200
Fixed selling costs	7,100

Required:

A. What is the variable cost per unit?

$ 15

B. What is the contribution margin per unit?

$ 34

C. What is the variable cost ratio?

31 %

D. What is the contribution margin ratio?

69 %

16. PR.07.165.ALGO (Algorithmic)

Fry Corporation developed the following income statement using a contribution margin approach:

Fry Corporation

Projected Income Statement For the Current Year

Sales (10,000 units)		$219,000
Less variable costs:		
Variable manufacturing costs	$69,500	
Variable selling costs	29,500	
Total variable costs		99,000
Contribution margin		$120,000
Less fixed costs:		
Fixed manufacturing costs	$90,000	
Fixed selling and administrative costs	27,000	
Total fixed costs		117,000
Operating income		$3,000

Required:

A. Determine the break-even point in sales dollars. **Round units Up to next whole unit. Use rounded amount in subsequent computations.**

$ ✓

213525

For B-C, round final answer to the nearest whole dollar.

B. The sales manager believed the company could increase sales by 250 units if advertising expenditures were increased by $13,000. By how much will the operating income increase or decrease if the advertising is increased as suggested?

$ ✓

-10000

decrease ✓

C. What is the maximum amount the company could pay for advertising if the advertising would increase sales by 250 units?

$ ✓

3000

Feedback

Check My Work

Correct

Post-Submission

A.

$117,000 ÷ ($120,000 ÷ $219,000) = $213,525

B.

(250 × $12.00) − $13,000 = $10,000 decrease

C.

250 × $12.00 = $3,000

Solution

Fry Corporation developed the following income statement using a contribution margin approach:

Fry Corporation

Projected Income Statement For the Current Year

Sales (10,000 units)		$219,000
Less variable costs:		
Variable manufacturing costs	$69,500	
Variable selling costs	29,500	
Total variable costs		99,000
Contribution margin		$120,000
Less fixed costs:		
Fixed manufacturing costs	$90,000	
Fixed selling and administrative costs	27,000	
Total fixed costs		117,000
Operating income		$3,000

Required:

A. Determine the break-even point in sales dollars. **Round units Up to next whole unit. Use rounded amount in subsequent computations.**

$ 213525

For B-C, round final answer to the nearest whole dollar.

B. The sales manager believed the company could increase sales by 250 units if advertising expenditures were increased by $13,000. By how much will the operating income increase or decrease if the advertising is increased as suggested?

$ -10000 decrease

C. What is the maximum amount the company could pay for advertising if the advertising would increase sales by 250 units?

$ 3000

17. PR.07.166.ALGO (Algorithmic)

Dan's Music plans to sell 5,100 MP3 players at $50 each in the coming year. Variable cost per unit is $10, and total fixed cost is $28,800.

Required:

For A-B, round calculation to 4 decimals, then enter as a percentage. (e.g., .298576, rounded to .2986, then entered as 29.86%)

A. Calculate the variable cost ratio.

✓

20.00

%

B. Calculate the contribution margin ratio.

✓

80.00

%

For C-D, round answer to the nearest whole dollar.

C. Calculate the break-even point in sales dollars.

$ ✓

36000

D. If Dan's Music has a target profit of $99,500, how many MP3 players will it have to sell?

✓

3208

MP3 players

Feedback

Check My Work

Correct

Post-Submission

A.

$10 ÷ $50 = 20%

B.

($50 − $10) ÷ $50 = 80%

C.

$28,800 ÷ 0.8 = $36,000

D.

($28,800 + $99,500) ÷ ($50 − $10) = 3,208 MP3 players

Solution

Dan's Music plans to sell 5,100 MP3 players at $50 each in the coming year. Variable cost per unit is $10, and total fixed cost is $28,800.

Required:

For A-B, round calculation to 4 decimals, then enter as a percentage. (e.g., .298576, rounded to .2986, then entered as 29.86%)

A. Calculate the variable cost ratio.

20.00 %

B. Calculate the contribution margin ratio.

80.00 %

For C-D, round answer to the nearest whole dollar.

C. Calculate the break-even point in sales dollars.

$ 36000

D. If Dan's Music has a target profit of $99,500, how many MP3 players will it have to sell?

3208 MP3 players

18. PR.07.170.ALGO (Algorithmic)

The following information was extracted from the accounting records of MVP Corporation:

Selling price per unit	$64
Variable cost per unit	4
Total fixed costs	528,000

Required:

A. What is MVP's break-even point in units?

✓

8800

units

B. How many units must be sold to earn an operating income of $60,000?

✓

9800

units

C. What is MVP's break-even point in units if the selling price increases by 23% and the variable costs decrease by 23%? **Round your intermediate calculations to 2 decimal places and round your final answer to next whole unit.**

✓

6981

units

D. Using the information in part (C), what sales level in dollars is needed to earn an operating income of $60,000? **Round your intermediate calculations to 2 decimal places and round your final answer to the nearest dollar.**

$ ✓

611969

Feedback

Check My Work

Correct

Post-Submission

A.

$528,000 ÷ ($64 per unit − $4 per unit) = 8,800 units

B.

($528,000 + $60,000) ÷ ($64 − $4) = 9,800 units

C.

($64 × 1.23) = $78.72 new sales price

($4 × 0.77) = $3.08 new variable cost

($78.72 − $3.08) = $75.64 new contribution margin

$528,000 ÷ $75.64 = 6,981 units

D.

Fixed costs + desired operating income ÷ contribution margin = unit sales needed

$528,000 + $60,000 = $588,000 ÷ ($78.72 − $3.08) = $75.64) = 7,774 units

7,774 units × $78.72 = $611,969 sales needed to earn operating income of $60,000

Solution

The following information was extracted from the accounting records of MVP Corporation:

Selling price per unit	$64
Variable cost per unit	4
Total fixed costs	528,000

Required:

A. What is MVP's break-even point in units?

8800 units

B. How many units must be sold to earn an operating income of $60,000?

9800 units

C. What is MVP's break-even point in units if the selling price increases by 23% and the variable costs decrease by 23%? **Round your intermediate calculations to 2 decimal places and round your final answer to next whole unit.**

6981 units

D. Using the information in part (C), what sales level in dollars is needed to earn an operating income of $60,000? **Round your intermediate calculations to 2 decimal places and round your final answer to the nearest dollar.**

$ 611969

19. PR.07.174.ALGO (Algorithmic)

Selina Company manufactures two products. Information about the two product lines for the year is as follows:

	Product X	Product Y
Selling price per unit	$65	$90
Variable costs per unit	33	46
Contribution margin per unit	$32	$44

The company expects fixed costs to be $124,000. The firm expects 60% of its sales (in units) to be of Product X.

Required:

Round units Up to next whole unit. Use rounded amount in subsequent computations.

Determine the break-even point in units for both Product X and Product Y.

✓

Product X: 2022 units

✓

Product Y: 1348 units

Feedback

Check My Work

Correct

Post-Submission

Form a package with 3 units of Product X and 2 units of Product Y.

Contribution margin from Product X: $32 × 0.60 = $19.20

Contribution margin from Product Y: $44 × 0.40 = $17.60

Total Package Contribution Margin = $19.20 + $17.60 = $36.80 per package

Break-Even Packages = $124,000 ÷ $36.80 per package = 3,370 packages

Product X: 3,370 packages × 0.60 = 2,022 units

Product Y: 3,370 packages × 0.40 = 1,348 units

Solution

Selina Company manufactures two products. Information about the two product lines for the year is as follows:

	Product X	**Product Y**
Selling price per unit	$65	$90
Variable costs per unit	33	46
Contribution margin per unit	$32	$44

The company expects fixed costs to be $124,000. The firm expects 60% of its sales (in units) to be of Product X.

Required:

Round units Up to next whole unit. Use rounded amount in subsequent computations.

Determine the break-even point in units for both Product X and Product Y.

Product X: 2022 units

Product Y: 1348 units

20. PR.07.176.ALGO (Algorithmic)

Newman Company expects to produce and sell 2,550 units next month. The data on costs are as follows:

Per-unit information:	
Selling price	$41
Variable manufacturing costs	11.00
Variable selling costs	6.25
Fixed costs:	
Fixed manufacturing costs	$12,800
Fixed selling costs	6,200

Required:

A. What is the break-even point in units? **Round units Up to next whole unit. Use rounded amount in subsequent computations.**

✓

800

units

B. What is the break-even point in sales dollars?

$ ✓

32800

C. What is the expected operating income for next month? **Round your answer to the nearest whole dollar.**

$ ✓

41563

D. What is the margin of safety in dollars?

$ ✓

71750

Feedback

Check My Work

Correct

Post-Submission

A.
Break-Even Units = ($12,800 + $6,200) ÷ $23.75 = 800 units

B.
Break-Even Sales Dollars = 800 × $41 = $32,800
OR
Break-Even Sales Dollars = $19,000 ÷ 0.579268 = $32,800

C.
Expected Operating Income = $104,550 − $43,988 − $19,000 = $41,563

D.
Margin of Safety = $104,550 − $32,800 = $71,750

Solution

Newman Company expects to produce and sell 2,550 units next month. The data on costs are as follows:

Per-unit information:	
Selling price	$41
Variable manufacturing costs	11.00
Variable selling costs	6.25
Fixed costs:	
Fixed manufacturing costs	$12,800
Fixed selling costs	6,200

Required:

A. What is the break-even point in units? **Round units Up to next whole unit. Use rounded amount in subsequent computations.**

800 units

B. What is the break-even point in sales dollars?

$ 32800

C. What is the expected operating income for next month? **Round your answer to the nearest whole dollar.**

$ 41563

D. What is the margin of safety in dollars?

$ 71750

21. PR.07.182.ALGO (Algorithmic)

Jacob Corporation had the following income statement for the current year:

Sales	$24,000
Variable expenses	15,000
Contribution margin	$9,000
Fixed expenses	3,000
Operating income	$6,000

Required:

A. Calculate the operating leverage ratio. **Round answer to three decimal places.**

✓

1.500

B. If sales increase by 17%, what will be the percentage change in income? **Round answer to two decimal places.**

✓

25.50

% increase ✓

C. If sales increase by $15,000, what will be the increase in income? **Round answer to nearest whole dollar.**

$ ✓

5,625

Feedback

Check My Work

Correct

Post-Submission

A.

$9,000 ÷ $6,000 = 1.5

B.

1.5 × 0.17 = 0.255, or 25.50% increase

C.

$15,000 × 0.375 = $5,625

OR 62.5% × 1.5 = 0.9375 and 0.9375 × $6,000 = $5,625

Jacob Corporation had the following income statement for the current year:

Sales	$24,000
Variable expenses	15,000
Contribution margin	$9,000
Fixed expenses	3,000
Operating income	$6,000

Required:

A. Calculate the operating leverage ratio. **Round answer to three decimal places.**

1.500

B. If sales increase by 17%, what will be the percentage change in income? **Round answer to two decimal places.**

25.50 % increase

C. If sales increase by $15,000, what will be the increase in income? **Round answer to nearest whole dollar.**

$ 5,625